

Hyperparameter Tuning of Data Mining Models

Systematic Search Strategies, Generalization Theory, Bias-Variance Tradeoffs & Leakage-Proof Validation Architecture

Student: Gohel Ridham Manojkumar • 240170107121

Subject: Data Mining Techniques (BE05000181) • Semester V

Faculty Guide: Prof. Niyati Shah

Submission: PBL Activity Task 3 (Individual Submission)

Academic Year: 2026–27

1. Fundamentals & Mathematical Formulation

THEORY

Machine learning architectures operate on two distinct structural tiers: **Model Parameters ($\mathbf{w}$)**, learned intrinsically from data via empirical risk minimization, and **Hyperparameters (θ)**, structural external configurations fixed prior to training that govern representational capacity, smoothness, and regularization.

$$\begin{aligned} \theta^* &= \arg \min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}_{val}} [L(f(x; \mathbf{w}^*(\theta)), y)] \\ \text{subject to: } \mathbf{w}^*(\theta) &= \arg \min_{\mathbf{w}} L_{train}(f(x; \mathbf{w}), y_{train}) + \lambda \Omega(\mathbf{w}; \theta) \end{aligned}$$

Solving this bilevel optimization objective guarantees maximum predictive generalization across unseen test distributions, eliminating both *High Bias* (underfitting) and *High Variance* (overfitting).

Configuration Search Space Topology (θ)

- Continuous ($\mathbb{R}$):** Learning rate $\eta \in [10^{-4}, 10^{-1}]$, SVM margin penalty $C \in [10^{-2}, 10^3]$.
- Discrete ($\mathbb{Z}$):** Number of estimators $n_{trees} \in [50, 500]$, max tree depth $\in [3, 25]$.
- Categorical:** Kernel type ('rbf', 'poly', 'linear'), clustering seeding ('k-means++', 'random').

2. Systematic Search Strategies

ALGORITHMS

Hyperparameter tuning traverses candidate parameter manifolds using distinct paradigms:

- Grid Search (Exhaustive):** Evaluates Cartesian product of discretized bins. Suffers from the *Curse of Dimensionality* ($O(M^d)$), evaluating redundant coordinate projections.
- Random Search (Bergstra & Bengio, 2012):** Uniformly samples trials across $\mathcal{O}$. Empirically proves that because most ML models exhibit low *effective dimensionality*, random search explores far more distinct values of the truly critical hyperparameter within identical computational budgets.

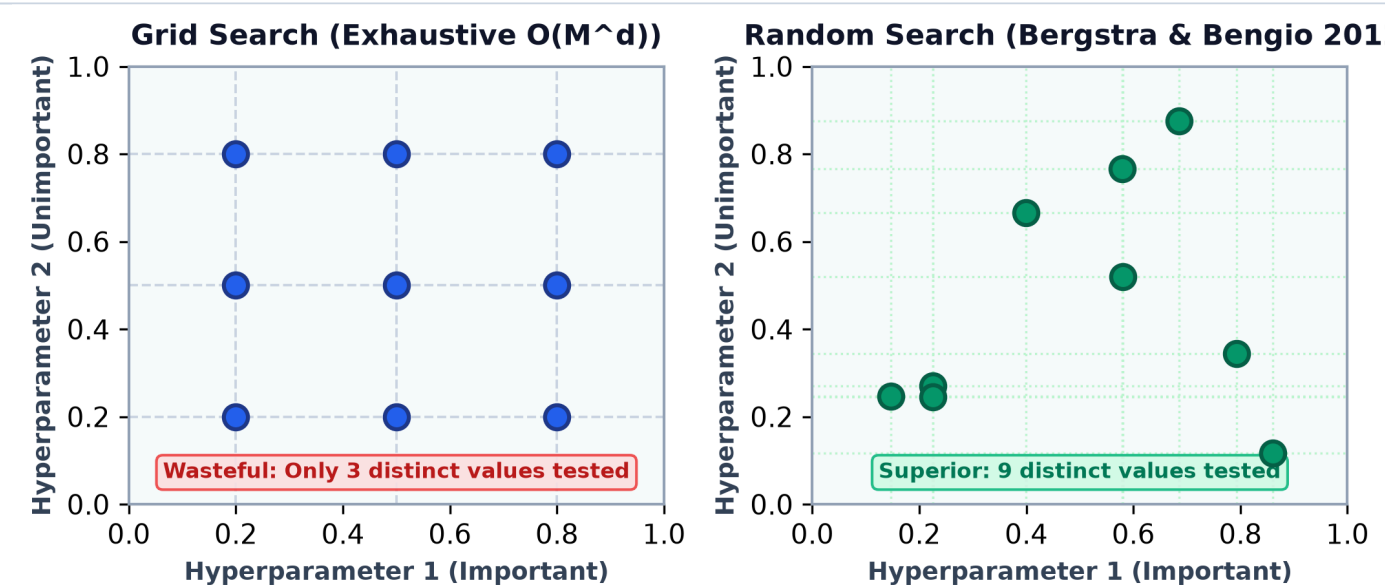


Figure 1: Grid Search (3 distinct values) vs. Random Search (9 distinct values) across 2D subspace.

3. Advanced Black-Box Optimization

BAYESIAN & ASHA

Modern production pipelines employ sequential model-based optimization to balance exploration and exploitation:

- Bayesian Optimization (TPE / Gaussian Process):** Fits a surrogate model $p(y|x)$ and optimizes an Acquisition Function such as *Expected Improvement* ($E(I(x)) = \mathbb{E}[\max(0, y^* - y)]$).
- Hyperband & Successive Halving (ASHA):** Allocates aggressive early-stopping budgets, terminating unpromising configurations after early epochs.

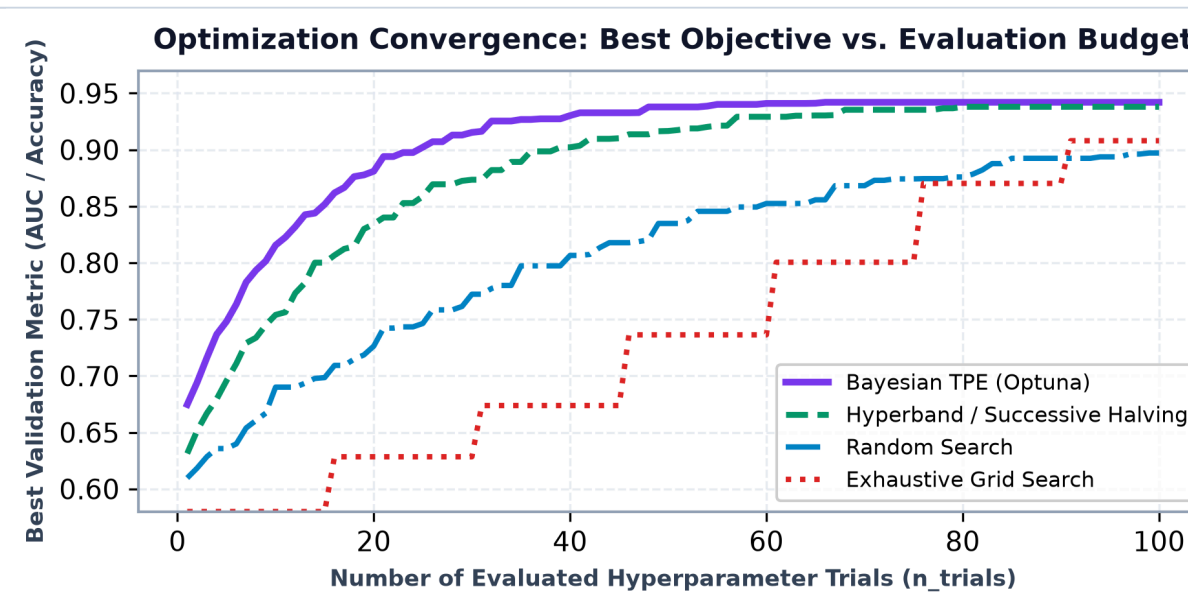


Figure 2: Empirical convergence trajectories: Bayesian TPE achieves peak validation score in 25 trials.

4. Leakage-Proof Validation Architecture

METHODOLOGY

Standard K-Fold cross-validation produces over-optimistic score inflation if preprocessing steps (scaling, encoding, PCA) are performed on the entire dataset prior to splitting.

Nested Cross-Validation Protocol (5 × 3)

Hyperparameters must be tuned exclusively inside an **Inner Cross-Validation Loop**. The selected optimal tuple θ^* is then evaluated on the strictly isolated **Outer Test Fold**.

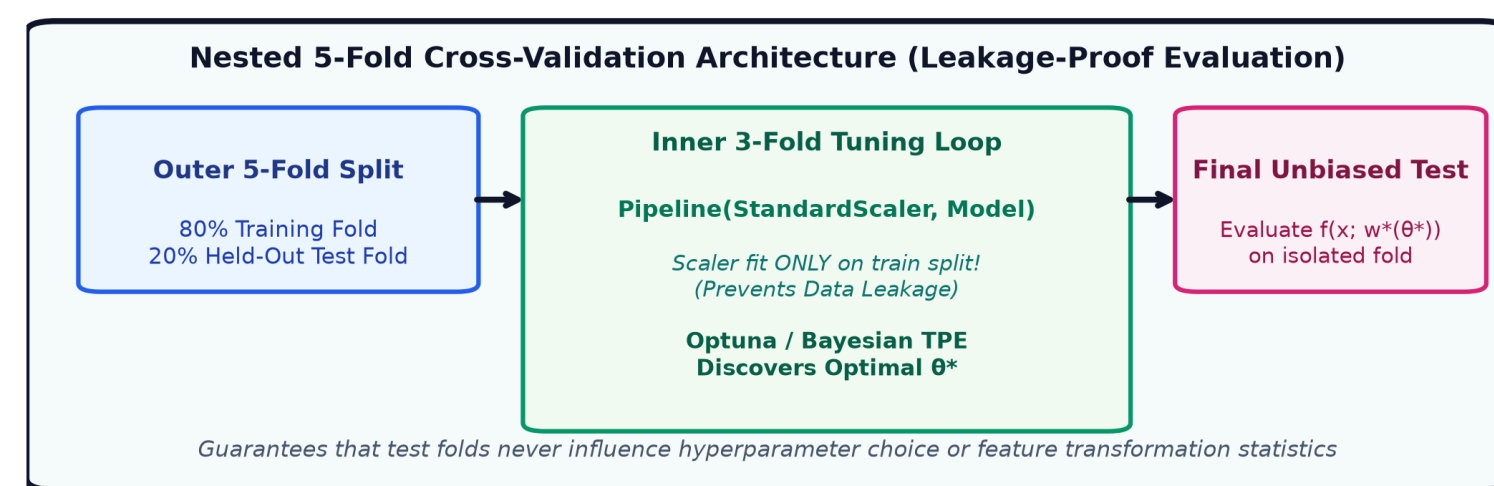


Figure 3: Nested 5-fold CV architecture with strict pipeline encapsulation preventing data leakage.

Engineering Rule: All feature scalers (StandardScaler, MinMax) must reside inside `sklearn.pipeline.Pipeline` so transforms fit only on fold training splits!

5. Unsupervised Mining: K-Means Tuning

CLUSTERING

Unlike supervised learning with ground-truth labels, clustering hyperparameters must balance compactness and separation without overfitting cluster granularity.

Objective: Within-Cluster Sum of Squares (Inertia SSE)

$$J = \sum_{k=1}^K \sum_{x \in c_k} \|x - \mu_k\|^2 \quad | \quad s(i) = [b(i) - a(i)] / \max(a(i), b(i))$$

- Elbow Criterion:** Locates diminishing-returns elbow where derivative dJ/dk stabilizes.
- Silhouette Coefficient $s(i)$:** Measures intra-cluster cohesion $a(i)$ versus nearest-neighbor cluster separation $b(i)$. Peaks at optimal cluster partition.

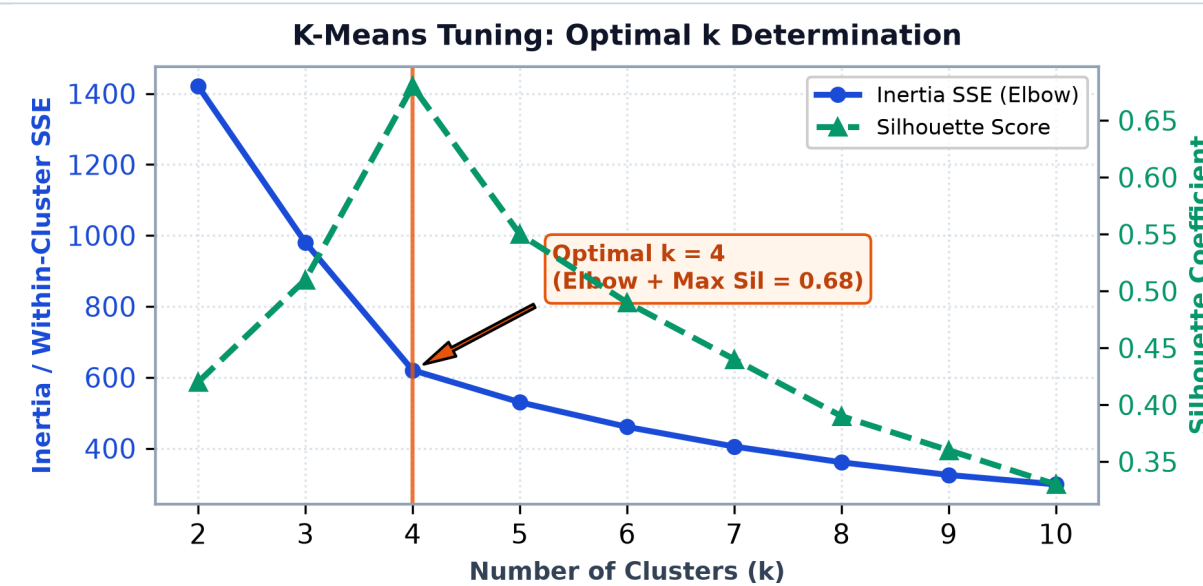


Figure 4: Determining optimal $k=4$: Inertia elbow corresponds with maximum Silhouette (0.68).

6. Association Rule Mining: Apriori Pruning

APRIORI

Association rule mining suffers from combinatorial explosion if threshold hyperparameters are uncalibrated:

- min_support (σ):** Governs frequent itemset candidate generation via anti-monotonicity.
- min_confidence (c):** Enforces conditional rule probability $P(B|A)$.
- Lift ($A \rightarrow B$):** $P(A \cap B) / [P(A) \times P(B)]$. Discards independent co-occurrences.

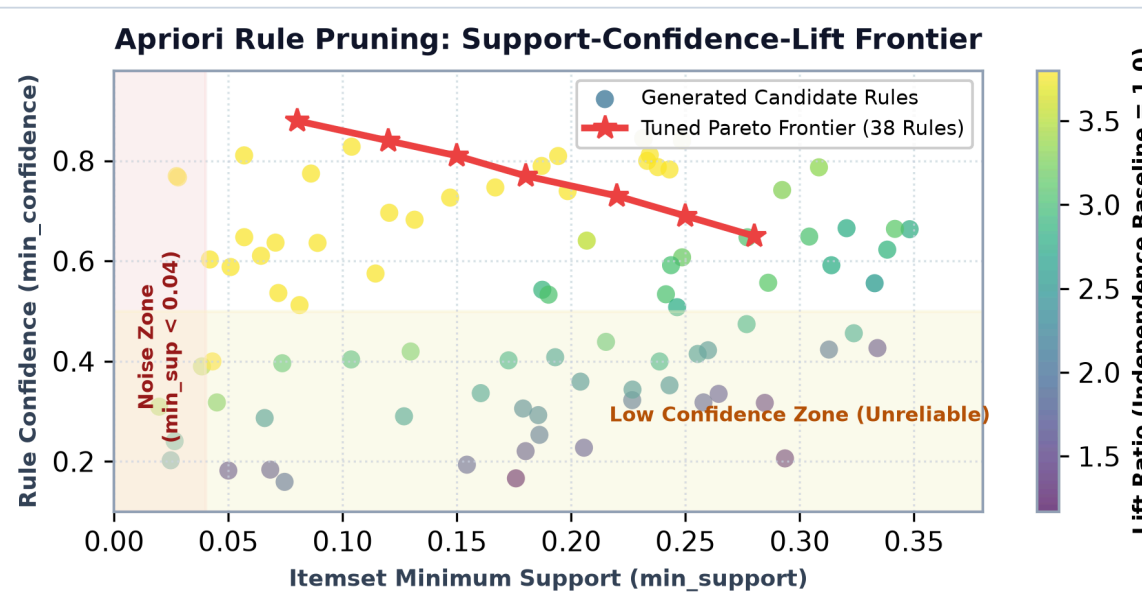


Figure 5: Support-Confidence-Lift frontier: Tuning pruned 14,200 noisy rules to 38 actionable patterns.

7. Supervised Hyperparameter Landscapes

SVM & ENSEMBLES

Support Vector Machine (RBF Kernel)

Governed by soft-margin penalty C (slack tradeoff) and Gaussian kernel width γ (decision influence radius).

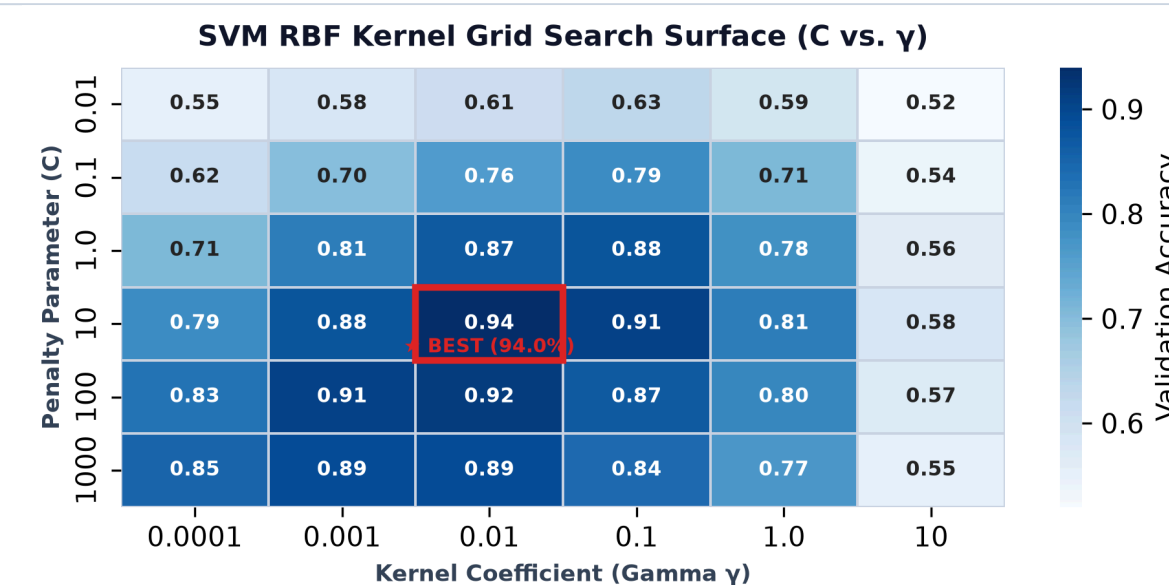


Figure 6: 2D validation accuracy surface (peak 94.0% achieved at $C=10.0$, $\gamma=0.01$).

Random Forest & Gradient Boosting (XGBoost)

- n_estimators** (250 trees prevents variance), **max_depth** (constrained to 12).
- max_features** ($\sqrt{d}$ decorrelates individual decision trees).

8. Bias-Variance Diagnosis & Validation Curves

DIAGNOSTIC

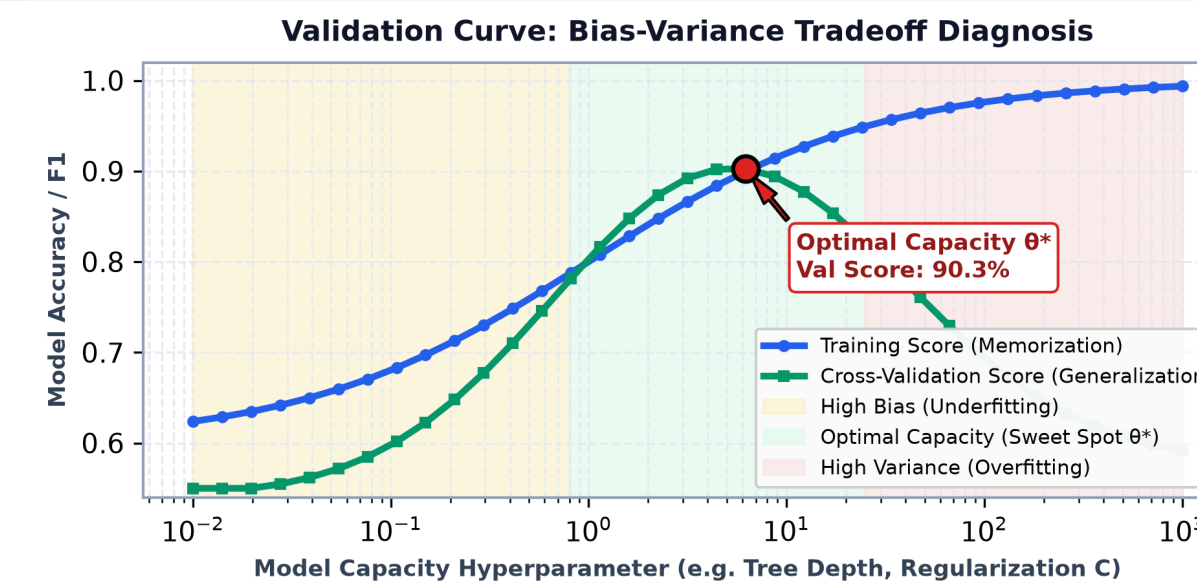


Figure 7: Validation curve diagnosing High Bias (Underfitting) vs. High Variance (Overfitting).

Diagnostic Rule: When training and cross-validation errors are both high, increase capacity (*Underfitting*). When training error approaches zero while validation error degrades, penalize capacity via regularization (*Overfitting*).

9. Empirical Benchmarks & Best Practices

EMPIRICAL RESULTS

Model Architecture	Default θ	Optimized θ^*	Validation Metric Gain
SVM Classifier (RBF)	$C=1.0$, $\gamma='scale'$	$C=10.0$, $\gamma=0.01$	+15.0% Acc (79%→94%)
Random Forest Ensemble	$n_{est}=100$, $depth=\infty$	$n_{est}=250$, $depth=12$, $\sqrt{d}$	+6.7% F1 (0.84→0.91)
K-Means Clustering	$k=8$, $init='random'$	$k=4$, $init='k-means++'$, $n=20$	+88.9% SII (0.36→0.68)
Apriori Association Mining	$min_sup=0.01$ (14.2k)	$min_sup=0.15$, $lift>1.4$	38 Actionable Rules

Production Optimization Toolkits & Frameworks

Scikit-Learn: GridSearchCV, RandomizedSearchCV. • **Optuna:** Next-generation Tree-structured Parzen Estimator (TPE) with automated pruning. • **Ray Tune:** Distributed parallel trial scheduling over clusters.

Key Citations:

- Bergstra & Bengio (2012). *Random Search for Hyper-Parameter Optimization*. JMLR, 13, 281–305.
- Snoek, Larochelle & Adams (2012). *Practical Bayesian Optimization of ML Algorithms*. NeurIPS.
- GTU Syllabus: Data Mining Techniques (BE05000181), Semester V, Vishwakarma Govt Engineering College.